

Open-Source Thermal Modeling for OpenROAD & OpenROAD Project Overview

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1. Open-Source Thermal Modeling Tools for Silicon Design

Key caveat: OpenROAD has no built-in thermal (temperature) solver. It handles power, timing, and IR-drop analysis (via OpenSTA and PDNSim), but temperature simulation always requires piping OpenROAD's power/floorplan data into a separate compact thermal simulator. "Plugging into OpenROAD" means using one of the bridge/flow projects below to feed one of the standalone solvers below.

1.1 Standalone Thermal Solvers (the physics engines)

HotSpot v7.0 — github.com/uvahotspot/HotSpot

- **Unique:** The de facto standard compact thermal model in academic IC/architecture research. Solves an equivalent RC thermal circuit from a floorplan (.flp) and per-block power trace. Fast, well-validated, supports 2D/3D stacks and microfluidic cooling; most other tools wrap it rather than reimplement it.
- **Limitations:** Compact/lumped model, not full finite-element analysis — coarser spatial resolution than real thermal FEA. Dated C codebase/CLI. No native OpenROAD hooks — you must generate the .flp/power files yourself from OpenROAD's DEF + power reports.

PACT (Parallel Compact Thermal Simulator) — github.com/peaclab/PACT

- **Unique:** Spans standard-cell-level to architecture-level granularity in one framework. Multi-core accelerated (OpenMPI). Natively models advanced cooling: liquid microchannels, 3D stacking, two-phase vapor chambers, thermoelectric coolers — not covered by HotSpot.
- **Limitations:** Default SuperLU solver cannot model liquid cooling or run in parallel (need the SPICE/Xyce solver path for that). Linux/macOS recommended; Docker on Windows is reportedly flaky. Steady-state and transient runs must share grid resolution.

3D-ICE — github.com/esl-epfl/3d-ice (EPFL project page)

- **Unique:** Purpose-built for vertically-stacked 3D-ICs with inter-tier microchannel liquid cooling — used in real hardware (IBM's Aquasar water-cooled server). ICCAD 2020 "10-year most influential paper" award winner. Supports heterogeneous/anisotropic floorplans and GDS-based layout input.
- **Limitations:** Linux-based, requires SuperLU MT setup. Narrower focus (liquid-cooled 3D stacks) than HotSpot/PACT's general-purpose scope. Standard open-source "no warranty" disclaimer — validate against your own data.

1.2 Bridges That Connect OpenROAD to a Thermal Solver

Open3DBench — github.com/lamda-bbo/Open3DBench (paper)

- **Unique:** Closest thing to a real "OpenROAD thermal plugin" today. Built directly on top of OpenROAD-flow-scripts: consumes DEF from its 3D placement stages, runs the OpenROAD backend (CTS, routing, RC extraction via OpenSTA/OpenRCX), generates per-cell power density from OpenSTA, and pipes it into HotSpot 7.0 for the thermal pass — a full PPA(+Thermal) loop.
- **Limitations:** Research benchmark, not a polished production tool — expect to read/adapt scripts. Documentation for the OpenROAD↔HotSpot glue is thin (spread across companion repos Place-MoL / OpenROAD-3D). Targets 3D/2.5D-IC use cases specifically — overkill for simple single-die 2D thermal maps.

CoMeT (successor to HotSniper) — github.com/marg-tools/CoMeT (paper)

- **Unique:** Full system-level toolchain — extends the Sniper performance simulator, computes power via McPAT/CACTI, and feeds HotSpot for interval thermal simulation of 2D/2.5D/3D core+memory systems (thermal-aware DVFS, task migration, reliability/wearout modeling included). MIT licensed.
- **Limitations:** Architecture-level (workload/microarchitecture simulation), not physical-design level — does not consume OpenROAD DEF/GDS directly. Fits system thermal behavior from an architectural power model, not post-layout power density. Inherits some HotSniper dependency weight (PinPlay, Docker).

Predecessor **HotSniper** (github.com/anujpathania/HotSniper) still exists but is explicitly superseded/subsumed by CoMeT.

1.3 Practical Recommendation

- **Post-layout thermal maps for a 2D design in OpenROAD:** extract per-instance power from an OpenSTA power report + placement coordinates from DEF, bin into a HotSpot .flp, run HotSpot directly (what Open3DBench automates).
- **3D-IC / chiplet work in OpenROAD:** use Open3DBench as the reference flow.
- **Liquid cooling / advanced cooling-tech modeling:** swap HotSpot for PACT or 3D-ICE as the backend solver in the same DEF→power→thermal pipeline.
- **Architecture/workload-level thermal (not post-layout):** use CoMeT.

2. OpenROAD Project Rundown

"OpenROAD" is really two things layered together: **the OpenROAD app** (a single C++/Tcl binary containing the place-and-route engine and its tool modules) and **OpenROAD-flow-scripts (ORFS)**, the orchestration repo that stitches OpenROAD together with external tools (Yosys for synthesis, KLayout/Magic for GDS streamout and DRC) into one autonomous RTL-to-GDSII flow, plus PDK setup and an autotuner for parameter search.

2.1 What It Has (Stable, In Daily Use)

Module	Full Name	Stage	Based On
sta / dbSTA	OpenSTA	Static timing / power analysis	OpenSTA
ifp	Floorplan init	Floorplanning	custom
pdn	PDN generation	Power grid synthesis	custom
tap	Tapcell / welltie insertion	Floorplanning	custom
ppl	I/O placement	Floorplanning	custom
mpl	Macro placer	Macro placement	Hier-RTLMP-derived
gpl	Global placer	Placement	RePIAce
dpl	Detailed placement	Placement legalization	OpenDP
rsz / est	Resizer / parasitics estimation	Timing repair, buffering	custom
cts	Clock tree synthesis	CTS	TritonCTS
grt	Global routing	Routing	FastRoute-derived
drt	Detailed routing	Routing	TritonRoute
ant	Antenna check / repair	Routing signoff	custom
rcx	OpenRCX	Parasitic extraction	custom

Module	Full Name	Stage	Based On
psm	PDN / IR-drop analysis (PDNSim)	Power signoff	custom
par	TritonPart	Partitioning	TritonPart
pad	ICeWall	Pad ring / chip-level I/O	custom
dft	Design-for-test	Scan insertion	custom
fin	Metal fill	Finishing	custom
gui	Qt-based GUI	Visualization / debug (all stages)	custom
odb	OpenDB	Shared physical database (LEF/DEF)	custom

Outside the core binary but part of the same ecosystem: **Yosys** (RTL synthesis, external project), **KLayout/Magic** (GDS streamout, DRC), **OpenLane/LibreLane** (a related flow wrapper on the same engines, from efabless), **OpenFASoC** (analog/mixed-signal generator on top of OpenROAD), and open PDKs like SkyWater 130nm, GF180MCU, and IHP SG13G2 that make the flow usable for real tapeouts (used in several Google/efabless MPW shuttle chips).

2.2 What's Actively Being Developed

- **Macro placement quality** — hierarchical/mixed-size placement (Hier-RTLMP lineage) is still an active research area; large-macro-count SoC floorplanning quality is a known weak spot vs. commercial tools.
- **Multi-corner/multi-mode STA and ECO flows** — OpenSTA supports multi-corner analysis, but the automated ECO loop (post-route timing fix → resize/buffer → re-route) is newer and still being hardened.
- **3D-IC / chiplet flows** — extensions for 2.5D/3D partitioning and place-and-route (e.g. Open3DBench) are frontier, not yet core/stable.
- **Machine-learning-assisted tuning** — the ORFS autotuner (Bayesian/RL parameter search for QoR) and ML-based routability/timing prediction are ongoing research threads, not fully productized.
- **GUI and scripting ergonomics** — the Python API surface is still catching up to the Tcl interface; GUI testing/coverage is an active 2025–2026 work item.
- **Detailed routing at scale/advanced nodes** — TritonRoute's DRC-clean convergence degrades past certain design-complexity/congestion thresholds; improving robustness here is ongoing.

2.3 What's Missing / Gaps vs. Commercial EDA

- **No native thermal solver** — temperature analysis requires bolting on an external tool (HotSpot, PACT, 3D-ICE) via a bridge flow (see Section 1); there is no first-party thermal engine.
- **No integrated signoff DRC/LVS** — signoff-grade DRC/LVS is delegated entirely to external tools (KLayout, Magic, Netgen); OpenROAD's own antenna/DRC checks are in-flow sanity checks, not foundry signoff.
- **Limited analog/mixed-signal and custom-layout support** — OpenROAD is digital-flow-first; analog is handled by adjacent, less mature projects (OpenFASoC) rather than the core tool.
- **QoR gap at advanced nodes** — published comparisons consistently show commercial P&R tools ahead on post-route PPA, especially as node complexity (multi-patterning, complex design rules) increases; the gap narrows steadily but hasn't closed.

- **No formal equivalence checking / formal verification** built in — left to separate tools (e.g. Yosys-based equivalence checking is bolted on, not integrated).
- **PDK coverage** — production-quality open PDKs are still limited to a handful of nodes (SkyWater 130nm, GF180, IHP 130nm); nothing near leading-edge commercial nodes, which caps what OpenROAD can be validated against.
- **Support model** — as an academic/nonprofit-led project (DARPA-seeded, UCSD-led, commercial backing via Precision Innovations), it lacks the SLA-backed support, certified PDK partnerships, and signoff certification that commercial EDA vendors provide.

Sources

- [HotSpot \(uvahotspot\)](#)
- [PACT \(peaclab\)](#)
- [3D-ICE \(esl-epfl\)](#)
- [3D-ICE project page \(EPFL\)](#)
- [Open3DBench \(lamda-bbo\)](#)
- [Open3DBench paper](#)
- [CoMeT \(marg-tools\)](#)
- [CoMeT paper](#)
- [HotSniper \(anujpathania\)](#)
- [OpenROAD PDNSim](#)
- [OpenROAD \(main repo\)](#)
- [OpenROAD-flow-scripts](#)
- [OpenROAD-flow-scripts docs](#)
- [OpenROAD DeveloperGuide \(tool namespace table\)](#)
- [OpenROAD documentation](#)
- [The OpenROAD Project site](#)
- [OpenSTA](#)
- [Kahng, "Looking Into the Mirror of Open Source"](#)
- [OpenROAD: Toward a Self-Driving, Open-Source Digital Layout Tool Chain](#)
- [Bridging Academic Open-Source EDA to Real-World Usability](#)